

DOE SBIR/STTR FY26 PHASE I • GENESIS MISSION

# Topic 1, Scaling the Biotechnology Revolution

Copy each answer into the matching open-text field in AMP. Word counts are shown so you can check them against the Pitch Development Guide before pasting. Images and video are not enabled, so all four answers are plain text.

|                                                  |                                           |                                                        |
|--------------------------------------------------|-------------------------------------------|--------------------------------------------------------|
| TOTAL WORDS<br><b>691</b><br>across four answers | QUESTION 1<br><b>97</b><br>held under 100 | TOPIC<br><b>Locked</b><br>after the pitch is submitted |
|--------------------------------------------------|-------------------------------------------|--------------------------------------------------------|

PROJECT TITLE TO ENTER IN AMP

Physics-Informed Machine Learning for Design of Continuous Immobilized-Enzyme Reactors  
Converting Post-Consumer PET to Fuel-Grade Ethanol

01

## Summary, Topic, and Mission Alignment

97 WORDS

PlastiBioFuel runs two loops: a structured wet lab converting post-consumer PET into fuel-grade ethanol, and a fast simulation loop that decides what the lab runs next. The chemistry is a PET hydrolase immobilized in a custom covalent organic framework inside a continuous spiral-flow reactor. Scale-up is limited by design search, not enzyme performance. Phase I builds a physics-informed machine learning model of that reactor, trains it on our own continuous-operation data, and closes the loop: the model proposes conditions, the reactor tests them, results retrain the model. That is AI for bioreactor design and biomanufacturing, Topic 1.

02

## Technical Promise

199 WORDS

**Significance.** Enzymatic PET recycling is limited by reactor engineering, not enzyme discovery. Operating conditions for continuous immobilized-enzyme reactors are still chosen by one-factor-at-a-time bench runs that take days each, so only a small part of the design space is ever searched.

**Innovation.** FAST-PETase was reported by the Alper group at UT Austin in 2022, itself produced by a machine learning method. PlastiBioFuel did not invent the enzyme. Our contributions, covered by USPTO provisional 63/848,456, are the covalent organic

framework immobilization system, the spiral-flow reactor geometry, and the integrated depolymerization-to-ethanol process. Phase I moves machine learning up one level, from residue selection to reactor operation: a physics-informed surrogate predicts conversion and turnover frequency from temperature, residence time, crystallinity, and enzyme loading, and an active-learning loop selects the next run.

**Feasibility.** The training data already exists. We have depolymerized real high-crystallinity post-consumer bottle PET to terephthalic acid, MHET, and ethylene glycol, confirmed by HPLC, loaded the framework to near-complete occupancy with activity retained, held product formation linear through 24 hours of continuous operation, and shown that turnover-frequency decline tracks substrate depletion rather than enzyme deactivation. Identifiable kinetics and a reactor that answers in hours make the active-learning loop tractable on a Phase I budget.

OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 34 WORDS

Phase I will produce a model whose predictions are tested against held-out reactor runs, so the go or no-go evidence at the end of Phase I is a measured prediction error, not a claim.

## 03

### Commercialization Potential

199 WORDS

**Value proposition.** The product is fuel-grade ethanol made from a waste feedstock. Post-consumer PET is abundant, domestically sourced, priced well below agricultural feedstock, and it does not compete with food production. Low carbon fuel standard credit value applies to the finished fuel and is the firmer part of the value case. Federal renewable identification number eligibility remains subject to final renewable-fuel pathway qualification, and we do not assume it.

**Competitive advantage.** Pyrolysis carries high energy cost per gallon; mechanical recycling degrades polymer on every cycle. Enzymatic recyclers that return monomer to virgin-equivalent resin sell into the resin market, so our offtake channel does not overlap theirs. AI-for-bioprocess software vendors have to contract for the asset we own: a working reactor generating the training data. Wet-lab groups search one condition at a time; we search at software speed. Owning both loops is what shortens the bench-to-plant step, where enzymatic routes usually stall.

**Go to market.** We are in offtake discussions with ARG Petro, a regional fuel distributor, and have their interest. Texas puts PET waste aggregation, refining infrastructure, and fuel blending demand in one region. The Phase I output is a sizing and operating envelope for the first modular unit, so the model feeds a capital decision.

OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 23 WORDS

A first modular unit sited near a Texas PET aggregation point is the specific commercial target that Phase I is meant to de-risk.

## 04

### Team Qualifications

196 WORDS

**Company.** Michael David Ward, Founder and CEO, is the principal investigator and is primarily employed by PlastiBioFuel. The company is a service-disabled veteran-owned small business verified through SBA VetCert and holds USPTO provisional 63/848,456. PlastiBioFuel runs both sides of this project in house: the structured laboratory program that produced the data cited here, and the simulation and model development that Phase I extends. All work will be performed in the United States.

**Partnerships.** The University of North Texas is our research partner under executed Sponsored Research Agreement SRA1016, with a \$250,000 extension under negotiation. Dr. Shengqian Ma, Department of Chemistry, leads covalent organic framework and immobilization chemistry; his group has published on enzyme immobilization in covalent organic frameworks and on framework host materials for enzymes. Dr. Joshua Phipps, postdoctoral researcher, runs day-to-day wet lab work. The partnership gives Phase I reactor access, HPLC analytics, and materials characterization with no capital spending.

**Follow-on funding.** PlastiBioFuel has a separate NSF SBIR Phase I proposal, Research.gov 328095, submitting against the November 4, 2026 window. That proposal covers wet chemistry and reactor hardware and does not overlap the computational scope proposed here. We disclose it so DOE can confirm there is no duplicate funding.

OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 19 WORDS

Marshall Nadel is a seed investor in PlastiBioFuel, and Patrick Tarlton of the Texas Concrete Association supports business development.

## WORD COUNT SUMMARY

As written · With add-on

|                                                  |                                        |
|--------------------------------------------------|----------------------------------------|
| <b>01. Summary, Topic, and Mission Alignment</b> | 97 words as written · 97 with add-on   |
| <b>02. Technical Promise</b>                     | 199 words as written · 233 with add-on |
| <b>03. Commercialization Potential</b>           | 199 words as written · 222 with add-on |
| <b>04. Team Qualifications</b>                   | 196 words as written · 215 with add-on |
| <b>Total</b>                                     | 691 words as written · 767 with add-on |

Question 1 is held under 100 words. If the Pitch Development Guide sets higher limits, append the add-on sentences. If it sets lower limits, the answers cut cleanly at paragraph boundaries. Drop the Feasibility paragraph from Question 2 last, since it carries the preliminary data that makes the pitch credible.

#### OPTIONAL UPLOAD: BIBLIOGRAPHY AND REFERENCES CITED

Prepared and included in this package. Worth uploading: it substantiates the statement that PlastiBioFuel did not invent the enzyme, and it shows the UNT partner's published record in enzyme immobilization in covalent organic frameworks.

PlastiBioFuel LLC · Michael David Ward, Founder and CEO ·  
michael@plastibiofuel.com · 817-319-7383

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